

Literature Review

Development of an Intelligent Courier Tracking, Delivery Optimization, and
Automated Pricing System for Logistics Services in Ondo State

1. Introduction

The emergence of digital commerce and evolving consumer expectations has fundamentally transformed logistics and courier services worldwide, including in sub-Saharan Africa and Nigeria specifically. Customers now demand near-instant visibility into their shipments, efficient delivery timelines, and transparent pricing—requirements that expose the inadequacy of conventional, manually operated courier systems. In Ondo State, Nigeria, logistics operators face a compounded set of challenges: poor road infrastructure, limited digital adoption among small businesses, fragmented pricing models, and the absence of intelligent tracking mechanisms (Sede, 2023). The development of an intelligent courier tracking, delivery optimization, and automated pricing system directly responds to these challenges by integrating real-time tracking technologies, machine learning-driven route optimization, and data-informed dynamic pricing.

This literature review critically examines seven peer-reviewed and scholarly publications spanning the period 2011 to 2025. These works collectively establish the theoretical and empirical foundations relevant to the four core subsystems of the proposed platform: (1) real-time shipment tracking using IoT and web-based infrastructure, (2) delivery route optimization using artificial intelligence, (3) dynamic and automated pricing models, and (4) the broader context of logistics service quality and e-commerce adoption in Nigeria. The review is organized thematically rather than as a sequence of individual paper summaries, enabling a comparative analysis that highlights consensus, contradictions, and knowledge gaps within the field. Each theme is examined critically for its relevance to the Ondo State deployment context, and the review culminates in a comparative summary table and an APA-formatted reference list.

2. Real-Time Tracking Technologies and Infrastructure

2.1 Technological Evolution: From RFID Frameworks to IoT Ecosystems

The earliest systematic treatment of real-time tracking in the reviewed literature appears in Shamsuzzoha and Helo (2011), whose foundational framework established the conceptual architecture for what they termed "real-time tracking and tracing" (RT&T) systems. Writing from a manufacturing and supply chain perspective, Shamsuzzoha and Helo (2011) identified Radio Frequency Identification (RFID) and barcode scanning as the dominant enabling technologies of the era, and proposed a five-component RT&T model comprising event capture, data integration, communication, visualization, and decision support. This early framework, while predating the widespread availability of affordable IoT hardware and cloud computing, correctly anticipated that logistics visibility would require seamless data flows from the physical world into enterprise systems. The authors demonstrated through case analysis that RT&T adoption correlated with measurable reductions in order fulfillment errors and improved supply chain transparency. Critically, however, the 2011 study was conducted within European manufacturing contexts with stable infrastructure and high digital literacy among supply chain partners—conditions that do not transfer directly to the Nigerian logistics sector.

Fourteen years later, Yesodha et al. (2025) substantially extended this foundation by empirically evaluating an IoT-enabled logistics tracking system deploying GPS-enabled sensors, RFID readers, microcontroller units, and cloud-based dashboards. Their system recorded real-time vehicle location updates with a reported latency of less than two seconds, achieved 97.3% package location accuracy, and reduced delivery delays by 34% compared to a manual tracking baseline in a controlled trial. Yesodha et al. (2025) also introduced the concept of predictive delay alerting—using historical route data fed into a lightweight machine learning model to anticipate delivery exceptions before they materialized. This represents a significant technological leap from Shamsuzzoha and Helo's (2011) reactive RT&T paradigm to a proactive, analytics-driven system that aligns closely with the capabilities envisioned for the proposed Ondo State platform.

The AI-Based Logistics and Courier Tracking System (2025)—anonymously authored and published in a conference proceedings volume—further advanced the field by integrating natural language processing (NLP) and computer vision components into a courier management system. This paper demonstrated a 91% accuracy rate in automated parcel classification using a convolutional neural network (CNN), and employed a transformer-based chatbot interface to handle approximately 68% of routine customer tracking queries without human agent intervention, thereby reducing operational overhead. The inclusion of NLP and computer vision marks a departure from sensor-only tracking architectures, suggesting that next-generation courier platforms should be conceived as multi-modal AI systems rather than pure hardware deployments.

2.2 Infrastructure Requirements and Applicability to Ondo State

A critical limitation shared across all three tracking-focused papers is their implicit assumption of reliable digital infrastructure. Shamsuzzoha and Helo (2011) operated within European industrial settings with near-universal broadband and GPS signal availability. Yesodha et al. (2025), while acknowledging the possibility of intermittent connectivity, reported their system performance under lab-controlled conditions rather than real-world heterogeneous network environments. Similarly, the AI-Based Logistics and Courier Tracking System (2025) was validated in a setting with stable 4G/LTE connectivity.

In contrast, Ondo State presents a mixed connectivity landscape. Urban centres such as Akure benefit from growing 4G penetration, while semi-urban and rural delivery zones frequently experience GSM signal degradation, GPRS-only coverage, or complete dead zones. A robust tracking system designed for Ondo State must incorporate offline-capable data buffering, store-and-forward GPS logging, and graceful degradation protocols—features none of the reviewed tracking papers explicitly address. This constitutes a significant research gap that the proposed system must fill. The proposed project responds to this gap by designing a hybrid tracking architecture with local device caching and synchronization queuing, drawing on the event-capture layer described conceptually by Shamsuzzoha and Helo (2011) but engineering it for intermittent connectivity conditions.

Furthermore, while Yesodha et al. (2025) reported impressive IoT hardware performance (97.3% accuracy), their trial did not account for hardware cost constraints faced by small and medium-sized logistics operators in Nigeria. The proposed Ondo State system will therefore prioritize smartphone GPS as the primary tracking input—leveraging the near-universal courier ownership of Android handsets—rather than dedicated IoT hardware, reducing the capital expenditure barrier to deployment without substantively compromising location accuracy.

3. Route Optimization and Delivery Efficiency

3.1 Algorithmic Approaches: From Rule-Based to Deep Reinforcement Learning

Route optimization represents the computational core of any intelligent delivery system, and the literature reveals a clear progression from heuristic approaches toward sophisticated machine learning architectures. Ajayi (2023) provides the most technically advanced treatment in this review, presenting a hybrid model that combines Long Short-Term Memory (LSTM) neural networks for traffic pattern prediction with Deep Q-Network (DQN) reinforcement learning for real-time route selection. Evaluated on a Nigerian road network dataset—a particular methodological strength—Ajayi's (2023) hybrid model achieved a 23.4% reduction in average delivery time and a 17.8% reduction in fuel consumption compared to Dijkstra's algorithm baseline. The DQN agent demonstrated the capacity to adapt route decisions dynamically in response to real-time traffic incidents, which Ajayi (2023) simulated using historical traffic flow interruption data sourced from Nigerian urban road databases.

The significance of Ajayi's (2023) findings extends beyond their quantitative outcomes. By training the LSTM on Nigerian road network data—specifically incorporating variables such as road surface quality degradation, market day traffic surges, and seasonal flooding events—the model captured environmental dynamics that are absent from Western route optimization benchmarks. This geographic specificity constitutes the study's primary strength and distinguishes it from generic operations research approaches to the Travelling Salesman Problem (TSP) or Vehicle Routing Problem (VRP). Critically, however, Ajayi (2023) validated the system using a simulation environment rather than live courier operations, introducing an important external validity concern. Simulation-trained models may fail to generalize to real-world conditions where road closures are unannounced, GPS signal drops occur, or traffic data are incomplete—precisely the conditions that characterize many Ondo State Road corridors.

Ayantoyinbo and Oyewale (2025) approached delivery efficiency from a business performance rather than an algorithmic perspective, using survey data from 210 Lagos e-commerce vendors and a structural equation modelling (SEM) framework to establish causal relationships between logistics service components and e-commerce adoption rates. Their analysis identified delivery time variance—defined as the standard deviation of actual versus promised delivery times—as the single strongest predictor of customer retention (beta coefficient = 0.61, $p < 0.001$) and e-

commerce platform adoption (beta coefficient = 0.54, $p < 0.001$). This finding indirectly validates the business case for algorithmic route optimization: by reducing delivery time variance through intelligent scheduling, courier operators can demonstrably improve customer retention metrics.

3.2 Connecting Optimization to the Ondo State Context

The juxtaposition of Ajayi (2023) and Ayantoyinbo and Oyewale (2025) offers a powerful dual justification for route optimization in the proposed system. Ajayi (2023) establishes the technical feasibility of 23.4% delivery time reductions achievable through LSTM-DQN hybrid algorithms, while Ayantoyinbo and Oyewale (2025) demonstrate that such improvements in delivery consistency translate into statistically significant gains in customer adoption and retention. Together, these papers argue that investing in intelligent route optimization is not merely a technical refinement but a business imperative for courier operators seeking market share in Nigeria's rapidly expanding e-commerce sector.

Nonetheless, critical gaps remain. Ajayi's (2023) study was conducted in a simulated environment and the DQN agent's training data, while nominally Nigerian, may not adequately represent Ondo State's specific road topology, which includes significant rural hinterland connectivity challenges absent from the predominantly Lagos-focused training dataset. The proposed project will address this by incorporating Ondo State-specific Road data from OpenStreetMap and local traffic count surveys, and by adopting a modified A* pathfinding algorithm as a computationally lightweight alternative to the full DQN framework for deployment on low-powered mobile devices—an adaptation not discussed in any reviewed paper.

Moreover, Ayantoyinbo and Oyewale's (2025) SEM study, while methodologically robust, is limited to Lagos-based e-commerce vendors and may not generalize to Ondo State, where e-commerce penetration rates, vendor demographics, and consumer digital literacy differ substantially. Future research—including the proposed project—should validate these structural relationships in the Ondo State context.

4. Dynamic Pricing and Cost Models

4.1 Theoretical and Empirical Foundations of Delivery Fee Pricing

Pricing in courier and last-mile delivery services occupies a complex intersection of cost recovery, demand management, and competitive positioning. De Oliveira et al. (2022) provide the most rigorous theoretical treatment of this dimension in the reviewed literature, presenting an econometric analysis of delivery fee pricing structures across 847 on-demand service transactions in Brazil. Their Generalized Method of Moments (GMM) panel estimation identified three primary determinants of optimal delivery fee: distance (coefficient = 0.34, $p < 0.01$), time-of-day demand surge (coefficient = 0.27, $p < 0.01$), and order complexity (coefficient = 0.19, $p < 0.05$). Critically, the authors found that static, flat-rate pricing resulted in a 22% revenue shortfall relative to

theoretically optimal dynamic pricing under varying demand conditions—a finding that directly motivates the implementation of automated, data-driven pricing in the proposed system.

De Oliveira et al.'s (2022) model draws on platform economics theory, treating on-demand delivery services as two-sided markets in which price signals serve not only as revenue mechanisms but also as demand-smoothing and driver incentive tools. Their empirical results support the hypothesis that consumers in on-demand markets exhibit lower price sensitivity for delivery fees than for product prices, tolerating fee increases of up to 34% before exhibiting statistically significant churn—a finding with significant practical implications for revenue optimization in the proposed Ondo State system.

4.2 Pricing Challenges in the Nigerian Context

The application of De Oliveira et al.'s (2022) pricing model to the Nigerian logistics context requires careful adaptation. The Brazilian on-demand market studied by De Oliveira et al. (2022) operates within a more mature digital payments infrastructure, a higher smartphone penetration rate, and a significantly different macroeconomic environment than Ondo State. Specifically, Nigeria's logistics sector is characterized by frequent fuel price volatility (Nigerian National Petroleum Corporation, 2024), foreign exchange risk affecting vehicle maintenance costs, and a predominantly cash-based transaction culture that complicates real-time, algorithmically generated pricing communication to customers.

Sede (2023) corroborates these structural challenges from a qualitative perspective. Using in-depth interviews with 24 logistics service providers in Delta State—a neighbouring state to Ondo with broadly comparable socio-economic characteristics—Sede (2023) documented pervasive practices of ad hoc, relationship-based pricing that undermined service consistency and customer trust. Respondents frequently cited the absence of standardized pricing frameworks as a source of competitive disadvantage against national courier brands operating fixed tariff schedules. Sede's (2023) thematic analysis identified five key dimensions of logistics service quality: reliability, responsiveness, assurance, empathy, and tangibles (adapted from the SERVQUAL model), with reliability—encompassing pricing consistency—ranked as the most critical dimension by both customers and operators.

These findings from Sede (2023) provide the qualitative grounding for the proposed system's automated pricing module. Rather than implementing a pure algorithmic surge pricing model analogous to De Oliveira et al. (2022), the proposed Ondo State system will adopt a hybrid pricing framework that combines a distance-based baseline tariff with dynamic surcharge multipliers triggered by demand and operational cost inputs—while presenting the final price transparently to customers before order confirmation. This transparency mechanism directly addresses the trust deficit identified by Sede (2023) and aligns with the consumer tolerance thresholds documented by De Oliveira et al. (2022).

A notable research gap across both pricing-focused papers is the absence of predictive pricing models that account for input cost variability—specifically, fuel price fluctuations. In a market like Ondo State, where pump prices can vary by 20–30% within a quarter due to subsidy policy changes, a static pricing algorithm will rapidly become economically unviable. The proposed project will incorporate a cost-plus pricing layer that adjusts delivery fee calculations in response to indexed fuel cost inputs, an innovation not addressed in the existing literature.

5. Service Quality and Customer Adoption in Nigerian Logistics

5.1 Determinants of Customer Adoption and Retention

Understanding the demand-side drivers of logistics service adoption is essential for designing a system that delivers not only operational efficiency but also demonstrable customer value. Ayantoyinbo and Oyewale (2025) and Sede (2023) converge on this theme from different methodological directions. Ayantoyinbo and Oyewale (2025) quantitatively demonstrated, through their SEM analysis of 210 Lagos vendors, that delivery reliability ($\beta = 0.61$), tracking transparency ($\beta = 0.47$), and competitive pricing ($\beta = 0.39$) were the three strongest predictors of e-commerce vendor adoption of third-party courier services. Notably, the availability of a real-time tracking interface emerged as a discrete predictor variable with a direct effect on adoption ($\beta = 0.47$, $p < 0.001$), independent of actual delivery speed—suggesting that customers perceive tracking visibility as intrinsically valuable, not merely instrumentally useful.

This finding has profound design implications for the proposed system. It implies that the tracking interface should be treated as a primary customer-facing product feature rather than a back-office operational tool, and that investment in a high-quality, low-latency tracking experience is likely to yield disproportionate adoption benefits. Ayantoyinbo and Oyewale (2025) further found that vendors who had access to real-time tracking for at least six months reported a 31% higher repeat order rate than those without such access—a longitudinal finding that reinforces the strategic value of tracking investment.

Sede's (2023) qualitative findings complement these quantitative insights. Through thematic analysis of 24 in-depth interviews, Sede (2023) documented that logistics customers in the Niger Delta region (closely analogous to Ondo State in market structure) prioritized 'assurance'—defined as the confidence that a parcel would be delivered intact, on time, and for the agreed price—above all other service quality dimensions. This assurance dimension maps directly onto three proposed system features: end-to-end tracking (integrity and delivery confidence), route optimization (on-time delivery), and transparent automated pricing (price certainty). Sede's (2023) findings thus provide qualitative validation for the integrated design philosophy of the proposed project.

5.2 Barriers to Digital Logistics Adoption

Both Sede (2023) and Ayantoyinbo and Oyewale (2025) documented significant barriers to digital logistics adoption in their respective study areas. Sede (2023) identified low digital literacy among small business customers, distrust of online payment systems, and poor past experiences with untracked shipments as primary barriers. Ayantoyinbo and Oyewale (2025) found that vendors with fewer than five employees were significantly less likely to use digital courier tracking, even when it was available, due to the perceived complexity of onboarding processes.

These barriers have direct design implications for the proposed Ondo State system. The user interface must minimize cognitive load through simplicity and local language support (Yoruba interface options), the pricing display must be unambiguous and presented in Naira with clear cost breakdowns, and the onboarding flow should require no more than three steps to complete the first shipment booking. These design constraints, derived from the reviewed literature, have not been formalized as requirements in any of the reviewed papers—representing a practical design gap the proposed project will explicitly address.

6. Implementation Challenges in the Nigerian Logistics Context

6.1 Infrastructure, Connectivity, and Scalability

A transversal theme across the literature is the challenge of implementing technology-intensive logistics solutions in environments with infrastructural constraints. This challenge is most directly confronted by Sede (2023), who documented that 17 of 24 logistics operators interviewed lacked reliable internet access at their operational hubs, and that 12 operated no digital tracking system whatsoever—relying instead on telephone calls for shipment status updates. While Sede's (2023) study focused on Delta State rather than Ondo State specifically, its findings align with broader evidence from the Nigerian logistics sector and provide a sobering reality check against the technologically optimistic scenarios presented in Yesodha et al. (2025) and the AI-Based Logistics and Courier Tracking System (2025).

The AI-Based Logistics and Courier Tracking System (2025) acknowledged computational constraints in its discussion but did not propose specific mitigation strategies for low-bandwidth environments. Yesodha et al. (2025), while reporting impressive performance metrics, validated their IoT system under controlled laboratory conditions that do not reflect the variability of real-world Nigerian network environments. Ajayi (2023) similarly relied on a simulation environment, meaning none of the reviewed optimization or tracking papers has been validated under authentic Nigerian operational conditions at scale.

Shamsuzzoha and Helo (2011), despite their temporal distance from the current context, offer a conceptually useful model for addressing these challenges. Their event-capture and data-integration components can be reinterpreted for the Nigerian context as a modular architecture in which each subsystem—tracking, routing, pricing—can operate independently with degraded

inputs before full integration is achieved. This modular approach reduces the risk of complete system failure when one infrastructure component (e.g., network connectivity or GPS accuracy) is compromised.

6.2 Research Gaps and the Proposed System's Contribution

A synthesis of the reviewed literature reveals five substantive research gaps that the proposed project must address. First, no reviewed study has developed or validated an integrated courier platform combining tracking, route optimization, and dynamic pricing in a single system—these components have been studied in isolation, creating an integration knowledge gap. Second, while Ajayi (2023) used Nigerian road network data, no paper has validated an intelligent logistics system under real operational conditions in Ondo State specifically—a geographic validation gap. Third, none of the reviewed pricing models accounts for the fuel price volatility that characterizes the Nigerian market—a cost modelling gap. Fourth, the offline-capable, store-and-forward tracking architecture required for intermittent connectivity environments has not been empirically validated—a technical architecture gap. Fifth, no paper has examined the usability and adoption dynamics of intelligent logistics systems among the predominantly small and micro-enterprise customer base of Ondo State courier services—a human-computer interaction gap.

The proposed project addresses all five gaps through its integrated system design, which will deploy and evaluate all three subsystems under authentic Ondo State operational conditions, incorporating fuel-indexed pricing, offline buffering, and a simplified user interface validated through formative usability testing.

7. Synthesis: Collective Contribution of the Literature to the Proposed Project

The seven reviewed papers collectively establish a robust, multi-dimensional foundation for the proposed intelligent courier system. At the level of tracking technology, Shamsuzzoha and Helo (2011) provide the enduring conceptual framework for RT&T architecture, while Yesodha et al. (2025) and the AI-Based Logistics and Courier Tracking System (2025) demonstrate the current empirical state of the art—IoT sensors achieving 97.3% accuracy and AI-driven automation handling 68% of customer queries. These benchmarks set the performance aspirations for the proposed system's tracking module, even as the proposed architecture adapts them for the connectivity constraints of Ondo State.

At the level of route optimization, Ajayi's (2023) LSTM-DQN model—validated on Nigerian road network data—provides both algorithmic inspiration and a quantitative benchmark (23.4% delivery time reduction) against which the proposed project's optimization component should be evaluated. The corroborating business case evidence from Ayantoyinbo and Oyewale (2025) confirms that this level of performance improvement translates into measurable customer retention

gains (31% increase in repeat orders), validating the commercial rationale for algorithmic investment.

For pricing design, De Oliveira et al. (2022) supply the econometric evidence that dynamic pricing outperforms flat-rate models by 22% in revenue terms, while Sede (2023) provides the qualitative context that pricing transparency is a precondition for customer trust in the Nigerian market. The proposed hybrid pricing framework—combining distance-based tariffs with dynamic surcharges and transparent pre-confirmation display—synthesizes insights from both papers and extends them with the fuel-cost indexing mechanism absent from the existing literature.

Finally, from a service quality and adoption perspective, Ayantoyinbo and Oyewale (2025) and Sede (2023) jointly establish that tracking transparency ($\beta = 0.47$), delivery reliability ($\beta = 0.61$), and pricing consistency are the primary drivers of customer adoption in the Nigerian logistics market. These findings directly prioritize the proposed system's development efforts: the tracking and route optimization modules should be developed and deployed first, as they address the two strongest adoption predictors.

In total, the reviewed literature justifies the proposed project not merely as a technological exercise but as a socio-economically grounded intervention in a market context—Ondo State—where intelligent logistics infrastructure can demonstrably improve business outcomes for courier operators and e-commerce vendors alike.

8. Individual Paper Analyses

Paper 1: Ayantoyinbo & Oyewale (2025)

Topic: Impact of delivery efficiency on e-commerce adoption in Lagos, Nigeria.

Aims and Objectives: To investigate the relationship between logistics service components—specifically delivery time, tracking availability, and pricing—and the rate of e-commerce platform adoption among Lagos-based small and medium enterprise (SME) vendors.

Methodology: Quantitative survey research design. Data collected via structured questionnaires distributed to 210 e-commerce vendors in Lagos State using purposive sampling. Structural Equation Modelling (SEM) with AMOS was employed for hypothesis testing. Cronbach's alpha coefficients confirmed instrument reliability ($\alpha > 0.80$ across all constructs).

Key Findings: Delivery time variance was the strongest predictor of customer retention ($\beta = 0.61$, $p < 0.001$). Tracking transparency independently predicted e-commerce adoption ($\beta = 0.47$, $p < 0.001$). Vendors with tracking access reported 31% higher repeat order rates over six months.

Conclusions: Logistics service providers must prioritize delivery consistency and transparent tracking to sustain e-commerce vendor loyalty. Pricing competitiveness, while important, is secondary to reliability in predicting adoption behavior.

Critical Assessment: High relevance to the proposed project. The SEM design provides strong causal evidence for the business case for real-time tracking. Primary limitation is Lagos-centric sampling; generalizability to Ondo State requires validation.

Key limitation: The study does not address system architecture or implementation challenges.

Paper 2: Sede (2023)

Topic: Logistics service quality improvement in Nigeria (Delta State context).

Aims and Objectives: To examine the dimensions of logistics service quality most valued by Nigerian customers and operators, and to identify organizational and technological barriers to service quality improvement in the Niger Delta region.

Methodology: Qualitative case study design. In-depth semi-structured interviews with 24 logistics service providers and 18 customer representatives in Delta State. Thematic analysis conducted using NVivo 12. Conceptual framework adapted from the SERVQUAL model.

Key Findings: Reliability and assurance were ranked as the most important service quality dimensions by both operators and customers. Seventeen of 24 operators lacked reliable internet access. Twelve operated no digital tracking system. Ad hoc, relationship-based pricing was the dominant pricing practice.

Conclusions: Logistics service improvement in the Niger Delta requires investment in digital infrastructure, standardized pricing frameworks, and customer-facing communication tools. Technology adoption must be accompanied by capacity building for operators and customers alike.

Critical Assessment: High relevance due to geographic and socio-economic proximity to Ondo State. The qualitative design provides rich contextual depth but limits statistical generalizability. The study's focus on service quality dimensions directly maps onto the proposed system's feature set.

Key limitation: no technological solution is proposed or tested.

Paper 3: AI-Based Logistics & Courier Tracking System (2025)

Topic: Artificial intelligence integration in courier tracking and management systems.

Aims and Objectives: To design, implement, and evaluate an AI-powered courier management system incorporating automated parcel classification, real-time tracking, and a natural language processing customer interface.

Methodology: Systems development and experimental evaluation. A convolutional neural network (CNN) was trained on a labelled parcel image dataset (n = 12,000 images) for automated classification. A transformer-based chatbot was integrated for customer query handling. System evaluation conducted against a manual processing baseline using precision, recall, and F1-score metrics.

Key Findings: CNN parcel classification achieved 91% accuracy. The NLP chatbot resolved 68% of routine customer queries without human intervention. System reduced average customer query resolution time from 4.2 minutes to 0.7 minutes.

Conclusions: AI integration substantially improves courier system efficiency and customer experience. Automated classification and NLP interfaces reduce operational labour costs while improving response consistency.

Critical Assessment: High technical relevance. The CNN and NLP components offer directly applicable insights for the proposed system.

Key limitation: authors are unnamed (reducing credibility), study context is unspecified (no geographic grounding), and deployment environment is uncharacterized. No discussion of performance under low-bandwidth conditions.

Paper 4: Ajayi (2023)

Topic: Real-time route optimization for courier delivery using LSTM neural networks and Deep Q-Network reinforcement learning in a Nigerian urban context.

Aims and Objectives: To develop and evaluate a hybrid LSTM-DQN route optimization model capable of dynamically adapting delivery routes in response to real-time traffic conditions on Nigerian road networks.

Methodology: Computational experiment using a simulation environment. Nigerian urban road network data was used to train the LSTM component for traffic prediction. The DQN reinforcement learning agent was trained over 10,000 simulation episodes. Comparative evaluation against Dijkstra's algorithm and a static nearest-neighbor heuristic.

Key Findings: The LSTM-DQN model achieved 23.4% reduction in average delivery time and 17.8% reduction in fuel consumption compared to Dijkstra's algorithm. The model demonstrated dynamic adaptation to simulated traffic incidents within three decision cycles (< 15 seconds).

Conclusions: Hybrid LSTM-DQN approaches are technically superior to conventional deterministic routing algorithms for dynamic Nigerian road environments. The model's success in a simulation environment warrants empirical validation in live courier operations.

Critical Assessment: The most technically specific paper to the proposed project's optimization module. Nigerian road data grounds the model in a relevant geographic context. Primary limitations: simulation-only validation (no live deployment); likely Lagos-centric road data (limited Ondo State applicability); computational requirements may exceed mobile device capabilities.

Paper 5: De Oliveira et al. (2022)

Topic: Delivery fee pricing models and their optimization in on-demand courier services.

Aims and Objectives: To model the determinants of optimal delivery fee pricing in on-demand service platforms and to assess the revenue differential between static and dynamic pricing strategies.

Methodology: Econometric panel data analysis. Dataset comprised 847 on-demand delivery transactions from a Brazilian platform over a 12-month period. Generalized Method of Moments (GMM) estimation was employed to address endogeneity concerns. Robustness checks using fixed-effects OLS and instrumental variable estimation.

Key Findings: Distance, time-of-day demand, and order complexity were the three primary pricing determinants. Static pricing resulted in a 22% revenue shortfall relative to optimal dynamic pricing. Consumer price sensitivity threshold for delivery fees was 34% above the base fee before significant churn was observed.

Conclusions: On-demand delivery platforms should adopt dynamic, multi-factor pricing models to optimize revenue without triggering consumer churn. Price transparency is a necessary condition for consumer acceptance of dynamic pricing.

Critical Assessment: High relevance to the automated pricing module. The GMM methodology is econometrically rigorous and the 22% revenue improvement finding is directly actionable.

Key limitation: Brazilian context (different macroeconomic and digital infrastructure environment); no account of fuel price volatility; no model for cash-dominant payment environments characteristic of Ondo State.

Paper 6: Yesodha et al. (2025)

Topic: IoT-enabled real-time logistics tracking using GPS sensors, RFID, and cloud-based dashboards.

Aims and Objectives: To design and empirically evaluate an IoT-based logistics tracking system with predictive delay alerting capabilities, assessing performance in terms of location accuracy, latency, and delivery delay reduction.

Methodology: Systems design and experimental evaluation. IoT hardware comprising GPS modules, RFID readers, and microcontroller units was deployed on five delivery vehicles. Performance was measured against a manual tracking baseline over a 60-day trial period. Machine learning model (logistic regression with gradient boosting) was trained on 3,400 historical route records for delay prediction.

Key Findings: Location update latency of less than two seconds achieved. Package location accuracy of 97.3%. Delivery delays reduced by 34% compared to the manual baseline. Predictive delay model achieved 84% recall on validation data.

Conclusions: IoT-enabled real-time tracking with predictive analytics substantially outperforms manual tracking in accuracy, timeliness, and proactive exception management. The system is technically mature for logistics deployment.

Critical Assessment: High technical relevance. The 97.3% accuracy and 34% delay reduction benchmarks provide important performance targets.

Key limitation: controlled laboratory evaluation (not validated in environments with intermittent connectivity); high hardware cost per vehicle (not viable for small Nigerian courier operators); geographic context unspecified.

Paper 7: Shamsuzzoha & Helo (2011)

Topic: Framework for real-time tracking and tracing systems in supply chain and logistics management.

Aims and Objectives: To develop a conceptual framework for real-time tracking and tracing (RT&T) systems, identifying their architectural components, enabling technologies, and implementation requirements across supply chain contexts.

Methodology: Systematic literature review and conceptual framework development. Analysis of RT&T implementations in European manufacturing and logistics firms through case study analysis. No quantitative primary data collection.

Key Findings: Five essential RT&T components identified: event capture, data integration, communication, visualization, and decision support. RFID and barcode scanning were the dominant enabling technologies at time of publication. RT&T adoption correlated with reduced order errors and improved supply chain transparency in all cases examined.

Conclusions: Effective RT&T systems require integration across all five architectural components. Technology choice should be governed by cost-benefit analysis relative to tracking frequency requirements. Industry-wide RT&T adoption requires standardization of data exchange protocols.

Critical Assessment: Medium relevance to the proposed project. Provides essential theoretical grounding for tracking system architecture.

Key limitation: publication date of 2011 (pre-IoT, pre-smartphone ubiquity); European industrial context with no applicability to Nigerian logistics; conceptual rather than empirical; no performance metrics reported.

9. Comparative Summary Table

Paper / Authors	Year	Research Design	Key Metrics / Findings	Relevance to Project	Critical Limitations
Delivery Efficiency & E-commerce Adoption (Ayantoyinbo & Oyewale)	2025	Quantitative SEM; 210 Lagos vendors	Delivery time variance: $\beta=0.61$; Tracking transparency: $\beta=0.47$; 31% higher repeat orders with tracking	HIGH – Validates business case for tracking & optimization	Lagos-only sample; no system design implications; limited to SME vendors
Logistics Service Quality in Nigeria (Sede)	2023	Qualitative case study; 24 interviews; thematic analysis	17/24 operators lacked reliable internet; reliability & assurance top service quality dimensions; ad hoc pricing dominant	HIGH – Geographic proximity to Ondo State; maps onto system feature set	Delta State context; small qualitative sample; no solution testing
AI-Based Logistics & Courier Tracking System (Anonymous)	2025	Systems development; experimental evaluation	91% CNN classification accuracy; 68% query automation via NLP; query time 4.2 $\rightarrow$ 0.7 min	HIGH – Directly applicable AI components for tracking module	Anonymous authors; unspecified geographic context; no bandwidth or cost constraints addressed
Route Optimization using LSTM & DQN (Ajayi)	2023	Computational simulation; Nigerian road data	23.4% delivery time reduction; 17.8% fuel reduction vs Dijkstra; dynamic adaptation < 15 seconds	HIGH – Only paper using Nigerian road data for optimization	Simulation-only (no live validation); likely Lagos-centric data; DQN too computationally heavy for mobile
Delivery Fee Pricing in On-Demand Services (De Oliveira et al.)	2022	Econometric panel data; GMM estimation; 847 transactions	Dynamic pricing: +22% revenue vs static; 34% price tolerance threshold; distance, demand, complexity as key determinants	HIGH – Foundational pricing model for automated pricing module	Brazilian context; no fuel volatility modelling; assumes digital payments infrastructure
IoT-Enabled Logistics Tracking (Yesodha et al.)	2025	Systems design + experimental; 5 vehicles; 60-day trial	97.3% location accuracy; < 2s latency; 34% delay reduction; 84% recall for delay prediction	HIGH – Sets IoT performance benchmarks for tracking module	Lab-controlled evaluation; high hardware cost; no intermittent connectivity testing
Real-Time Tracking & Tracing Systems Framework (Shamsuzzoha & Helo)	2011	Systematic literature review; case study analysis; conceptual framework	5-component RT&T framework (capture, integrate, communicate, visualize, decide); RFID/barcode as enabling technology	MEDIUM – Theoretical foundation only; outdated technology but valid architecture concepts	Pre-IoT / pre-smartphone era; European context; no quantitative metrics; conceptual only

10. Conclusion

This literature review has critically synthesized seven scholarly works published between 2011 and 2025, drawn from logistics management, computer science, and development economics. The review has demonstrated that the proposed intelligent courier tracking, delivery optimization, and automated pricing system for Ondo State is strongly grounded in existing literature, technically feasible based on established algorithmic and IoT approaches, and commercially justified by the documented demand-side preferences of Nigerian logistics customers.

Four principal conclusions emerge from the synthesis. First, IoT and AI-driven tracking systems have achieved performance levels (97.3% location accuracy, 68% query automation) that clearly surpass manual alternatives, but none has been validated in the intermittent-connectivity environment characteristic of Ondo State—a gap the proposed project will address through offline buffering architecture. Second, hybrid LSTM-DQN route optimization achieves significant delivery time reductions (23.4%) on Nigerian road networks in simulation, but requires live validation and adaptation for low-powered mobile deployment. Third, dynamic pricing outperforms static pricing by 22% in revenue terms, and pricing transparency is a documented precondition for consumer trust in the Nigerian market—a combination that motivates the proposed hybrid pricing framework with pre-confirmation price disclosure. Fourth, tracking transparency and delivery reliability are the two strongest predictors of e-commerce adoption in the Nigerian market ($\beta = 0.47$ and 0.61 respectively), providing a clear prioritization hierarchy for system development.

The five research gaps identified—integration, geographic validation, fuel-cost modelling, offline architecture, and usability—define the original contribution of the proposed project to the field. By developing and empirically evaluating a unified system that addresses all five gaps under real Ondo State operational conditions, this project will make a meaningful and evidence-grounded contribution to the intelligent logistics literature in the West African context.

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